

Linear Algebra Practice

1. Use the following code to read in data and store the results in `Credit`

```
Credit <- read.csv("http://statlearning.com/s/Credit.csv")
```

2. Create a linear model object named `mod_num` by regressing `Balance` onto `Income`, `Limit`, `Rating`, `Cards` and `Age`.

```
mod_num <- lm(Balance ~ Income + Limit + Rating + Cards + Age, data = Credit)
mod_num
```

Call:

```
lm(formula = Balance ~ Income + Limit + Rating + Cards + Age,
    data = Credit)
```

Coefficients:

(Intercept)	Income	Limit	Rating	Cards	Age
-449.3610	-7.5621	0.1286	2.0224	11.5527	-0.8883

```
summary(mod_num)
```

Call:

```
lm(formula = Balance ~ Income + Limit + Rating + Cards + Age,
    data = Credit)
```

Residuals:

Min	1Q	Median	3Q	Max
-231.37	-113.46	-39.55	41.66	544.35

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-449.36101	40.57409	-11.075	<2e-16 ***
Income	-7.56211	0.38214	-19.789	<2e-16 ***
Limit	0.12855	0.05289	2.430	0.0155 *
Rating	2.02240	0.79208	2.553	0.0110 *
Cards	11.55272	7.06285	1.636	0.1027
Age	-0.88832	0.47781	-1.859	0.0638 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 161.6 on 394 degrees of freedom

Multiple R-squared: 0.8781, Adjusted R-squared: 0.8765

F-statistic: 567.4 on 5 and 394 DF, p-value: < 2.2e-16

3. Create the model matrix and store the results in X.

```
X <- model.matrix(mod_num)
head(X)
```

	(Intercept)	Income	Limit	Rating	Cards	Age
1	1	14.891	3606	283	2	34
2	1	106.025	6645	483	3	82
3	1	104.593	7075	514	4	71
4	1	148.924	9504	681	3	36
5	1	55.882	4897	357	2	68
6	1	80.180	8047	569	4	77

4. Create the matrix $X'X$ and store the result in XTX.

```
XTX <- t(X)%*%X
XTX
```

	(Intercept)	Income	Limit	Rating	Cards	Age
(Intercept)	400.00	18087.55	1894240	141976	1183.00	22267
Income	18087.55	1313520.38	111365789	8141878	53141.58	1049422
Limit	1894240.00	111365788.86	11096147930	814393420	5615136.00	107050369
Rating	141976.00	8141878.10	814393420	59944846	424401.00	8013311
Cards	1183.00	53141.58	5615136	424401	4249.00	66260
Age	22267.00	1049421.59	107050369	8013311	66260.00	1358273

5. Find the inverse of $(X'X)^{-1}$ and store the result in `XTXI`.

```
XTXI <- solve(XTX)
XTXI
```

	(Intercept)	Income	Limit	Rating
(Intercept)	6.306995e-02	1.324119e-04	3.601959e-05	-5.911536e-04
Income	1.324119e-04	5.594479e-06	-1.203368e-08	-8.207053e-07
Limit	3.601959e-05	-1.203368e-08	1.071806e-07	-1.595510e-06
Rating	-5.911536e-04	-8.207053e-07	-1.595510e-06	2.403587e-05
Cards	-7.040877e-04	8.364838e-06	7.728855e-06	-1.172889e-04
Age	-4.531446e-04	-1.110866e-06	7.410214e-09	-8.034455e-09

	Cards	Age
(Intercept)	-7.040877e-04	-4.531446e-04
Income	8.364838e-06	-1.110866e-06
Limit	7.728855e-06	7.410214e-09
Rating	-1.172889e-04	-8.034455e-09
Cards	1.911105e-03	-5.326415e-06
Age	-5.326415e-06	8.746389e-06

6. Verify that the answer you computed in 5. is the same thing as `summary(mod_num)$cov.unscaled`.

```
summary(mod_num)$cov.unscaled -> p6
p6
```

	(Intercept)	Income	Limit	Rating
(Intercept)	6.306995e-02	1.324119e-04	3.601959e-05	-5.911536e-04
Income	1.324119e-04	5.594479e-06	-1.203368e-08	-8.207053e-07
Limit	3.601959e-05	-1.203368e-08	1.071806e-07	-1.595510e-06
Rating	-5.911536e-04	-8.207053e-07	-1.595510e-06	2.403587e-05
Cards	-7.040877e-04	8.364838e-06	7.728855e-06	-1.172889e-04
Age	-4.531446e-04	-1.110866e-06	7.410214e-09	-8.034455e-09

	Cards	Age
(Intercept)	-7.040877e-04	-4.531446e-04
Income	8.364838e-06	-1.110866e-06
Limit	7.728855e-06	7.410214e-09
Rating	-1.172889e-04	-8.034455e-09
Cards	1.911105e-03	-5.326415e-06
Age	-5.326415e-06	8.746389e-06

```
all.equal(XTXI, p6, tolerance = 0.0000001)
```

```
[1] TRUE
```

7. Use the normal equations to find the coefficients for `mod_num`. Verify that your estimates computed with the normal equations are the same as `coef(mod_num)`.

```
betahat <- XTXI%*%t(X)%*%Credit$Balance  
betahat
```

```
          [,1]  
(Intercept) -449.3610111  
Income       -7.5621098  
Limit        0.1285533  
Rating       2.0224047  
Cards       11.5527211  
Age        -0.8883161
```

```
coef(mod_num)
```

(Intercept)	Income	Limit	Rating	Cards	Age
-449.3610111	-7.5621098	0.1285533	2.0224047	11.5527211	-0.8883161

8. What is the MSE for `mod_num`.

```
MSE <- summary(mod_num)$sigma^2  
MSE
```

```
[1] 26102.08
```

9. Compute the variance covariance matrix for your estimated coefficients. Store the values in an object named `var_cov_b`.

```
var_cov_b <- MSE*XTXI  
var_cov_b
```

	(Intercept)	Income	Limit	Rating	Cards
(Intercept)	1646.2566016	3.4562250646	0.9401860793	-15.430336921	-18.3781508
Income	3.4562251	0.1460275269	-0.0003141041	-0.021422113	0.2183396
Limit	0.9401861	-0.0003141041	0.0027976350	-0.041646121	0.2017392
Rating	-15.4303369	-0.0214221133	-0.0416461205	0.627386036	-3.0614829
Cards	-18.3781508	0.2183396330	0.2017391560	-3.061482918	49.8838057
Age	-11.8280160	-0.0289959190	0.0001934220	-0.000209716	-0.1390305

	Age
(Intercept)	-11.828016032
Income	-0.028995919
Limit	0.000193422
Rating	-0.000209716
Cards	-0.139030492
Age	0.228298906

10. Compute the standard errors for your estimated coefficients. Verify that your answers are the same as `summary(mod_num)$coef[, 2]`.

```
se_b <- diag(var_cov_b)^.5
se_b
```

	Income	Limit	Rating	Cards	Age
(Intercept)	40.57408781	0.38213548	0.05289267	0.79207704	7.06284686

```
summary(mod_num)$coef[,2]
```

	Income	Limit	Rating	Cards	Age
(Intercept)	40.57408781	0.38213548	0.05289267	0.79207704	7.06284686

11. Compute the t values using your answers from 7. and 10. Verify that the values are the same as `summary(mod_num)$coef[, 3]`

```
Ts <- betahat/se_b
Ts
```

	[,1]
(Intercept)	-11.075074
Income	-19.789080
Limit	2.430456
Rating	2.553293
Cards	1.635703
Age	-1.859155

```
summary(mod_num)$coef[, 3]
```

(Intercept)	Income	Limit	Rating	Cards	Age
-11.075074	-19.789080	2.430456	2.553293	1.635703	-1.859155

12. Compute the p-values for testing $H_0 : \beta_i = 0$ versus $H_0 : \beta_i \neq 0$ for $i = 0, \dots, 5$.

```
ps <- pt(abs(Ts), 394, lower = FALSE)*2
ps
```

```
      [,1]
(Intercept) 5.303629e-25
Income      5.143084e-61
Limit       1.552541e-02
Rating      1.104684e-02
Cards       1.027003e-01
Age         6.375045e-02
```

```
summary(mod_num)$coef[, 4]
```

(Intercept)	Income	Limit	Rating	Cards	Age
5.303629e-25	5.143084e-61	1.552541e-02	1.104684e-02	1.027003e-01	6.375045e-02